

AI ENGINEER CAREER ROADMAP

Student → AI Engineer | Timeline: 12 Months

CURRENT POSITION

Role: Student

Skills: Python · Web Development

Level: Beginner → Intermediate

Strength: Coding foundation already in place

TARGET GOAL

Role: AI / ML Engineer

Timeline: 12 Months

Avg Salary: \$95K–\$150K (entry level)

Demand: Very High — fastest growing tech role

SKILL GAP ANALYSIS

SKILL AREA	YOU HAVE	GAP	PRIORITY
Machine Learning	Basics only	Algorithms, scikit-learn, model eval	HIGH
Deep Learning	None	PyTorch / TensorFlow, NNs, CNNs	HIGH
LLMs & Prompt Eng.	None	Transformers, LangChain, RAG, Fine-tuning	HIGH
MLOps & Deployment	None	Docker, FastAPI, HuggingFace, AWS/GCP	MED
Math (Lin. Algebra/Stats)	Partial	Probability, matrices, backprop intuition	MED
Python for ML	Present	NumPy, Pandas, Matplotlib polish	LOW
Web Dev (APIs)	Present	FastAPI integration for model serving	LOW

LEARNING PLAN & SUGGESTED PROJECTS

PHASE	MONTH	FOCUS AREA
1 — Foundation	1–3	ML fundamentals · NumPy · Pandas · scikit-learn · Math refresher
2 — Deep Learning	4–6	PyTorch basics · CNNs · RNNs · Hugging Face Transformers
3 — LLMs & GenAI	7–9	Prompt Engineering · LangChain · RAG pipelines · Fine-tuning
4 — Deploy & Ship	10–12	FastAPI · Docker · Cloud (AWS/GCP) · GitHub portfolio · Job apps

SUGGESTED PROJECTS

- Movie Sentiment Analyzer — scikit-learn NLP (Month 2)
- Image Classifier App — PyTorch + Streamlit UI (Month 5)
- Personal AI Chatbot — LangChain + RAG + your docs (Month 8)
- Resume Screener API — FastAPI + LLM + Docker deploy (Month 11)
- Open-Source Contribution — HuggingFace / LangChain (Month 12)

NETWORKING STRATEGY

LinkedIn: Post weekly AI learnings + project demos

GitHub: Push code daily — green squares matter!

Twitter/X: Follow Andrej Karpathy, Yann LeCun, @huggingface

Community: Join fast.ai forums, Kaggle, Discord AI servers

Events: Attend local ML meetups / virtual AI conferences

Reach out: DM 2 AI engineers/month for informational chats

MONTHLY MILESTONES

M1-3: Complete ML course · 2 mini projects on GitHub

M4-6: Build deep learning project · 500 LinkedIn followers

M7-9: Deploy RAG chatbot · Start applying for internships

M10: 4+ portfolio projects · Resume polished · Referrals

M11: Active job applications (20+/week) · Mock interviews

M12: Offer in hand or internship converting to full-time ■

■ IMMEDIATE NEXT ACTIONS — START THIS WEEK

- | | |
|---|---|
| 1 | Enroll in fast.ai 'Practical Deep Learning' (free) or Andrew Ng's ML Specialization on Coursera |
| 2 | Create a GitHub profile — commit your first ML notebook today |
| 3 | Set up your LinkedIn headline: 'CS Student Aspiring AI Engineer Python · Web Dev' |
| 4 | Install PyTorch + Jupyter locally — run your first tensor operation |
| 5 | Bookmark: HuggingFace.co · Papers With Code · Kaggle.com |

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